CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANGA

M.Tech. (EC)(C.S.E.) Sem - I Examination, 2009-10
EC – 702 Digital Communication & Coding

Date: 23.12.09 , Wednesday **Time: 01:00 p.m to 04:00 p.m** **Maximum Marks:70**

Instructions:
1. Attempt all questions.
2. Answer to the two sections must be written in the separate answer books.
3. Figures to the right indicate *full* marks.
4. Make suitable assumptions and draw the neat figures wherever required.

SECTION- I

Q.1

A The PDF of two continuous RVs is given by 3

$$P_{xy}(x,y) = x\, y\, e^{-(x2+y2)/2}\, u(x)\, u(y)$$

a) Find $P_x(x)$, $P_y(y)$, $P_{x/y}(x/y)$ and $P_{y/x}(y/x)$
b) Are x and y independent?

B State and explain Bayes' Theorem 3

C "An ergodic process is stationary but a stationary process is not necessarily ergodic". Comment 3

D The PDF of amplitude x of a certain signal x(t) is given by 3

$$P_x(X) = 0.5\ |x|\ e^{-|x|}$$

Find a) Probability that $x \geq 1$
b) Probability that $-1 < x \leq 2$
c) Probability that $x \leq 2$

Q.2

A Give any one example of non-linear modulation with memory explain the same. 6

B A speech signal is sampled at a rate of 8 KHz, logarithmically compressed and encoded into a PCM format using 8 bits per sample. The PCM data is transmitted through an AWGN channel via M-level PAM. Determine the bandwidth required for transmission when 5

i) M=4 ii) M=8 iii) M=16

OR

Q. 2

A Explain correlation type demodulator. 6

B Explain the Costas loop for generating a properly phased carrier for DSB-SC 5

Q.3

A Write short note on Carrier recovery with decision feedback PLL. 6

B State and prove Nyquist condition for zero ISI. 6

OR

Q.3

A Explain the effect of ISI on eye opening. 6

B Explain the representation of Band pass signal. 6

SECTION – II

Q4

A A zero-memory source emits messages m_1 and m_2 with probabilities 0.8 and 0.2 respectively. Find the optimum (Huffman) binary code for this source as well as for its second and third order extensions. Determine the code efficiencies in each case. 6

B Design the code tree and state diagram for coder given as $N=3$ & $\upsilon=2$ 6

where $\upsilon_1 = s_1 \oplus s_2 \oplus s_3$,

$\upsilon_2 = s_1 \oplus s_3$

Q.5

A Find the generator polynomial for g(x) for a (7,4) cyclic code and find code vector for the following data vector 1111, 0000, 1001, 1100. 6

B Derive the expression for the channel capacity of a non-ideal filter channel. 5

OR

.5

A "The channel capacity of continuous AWGN channel always remains finite for finite signal and noise powers." Justify. 5

B Describe the FFT based multicarrier system. 6

Q.6 Write short notes on 12

i) Viterbi Algorithm

ii) LMS Algorithm

OR

Q.6 Write short notes on 12

i) Tap-leakage Algorithm

ii) Predictive decision feedback equalizer